

Q8) Give note on low cost excreta disposal systems. [5]

- Q9)** a) Explain the One-Pipe System and Two pipe system in detail. [7]
b) What are the factors to be consider for design of Water supply in buildings? Discuss. [7]

OR

- Q10)** a) Give note on Break pressure tank system and Hydro-pneumatic system. [7]
b) Discuss the steps involved in Design of Water pipes. [7]

- Q11)** a) Discuss with suitable sketches about Gully Traps and Intercepting Traps. [8]
b) Discuss the working of Flushing Cistern, Wash Basin, Lavatory Basin. [8]

OR

- Q12)** a) Give detail note on principles governing design of building drainage. [8]
b) Discuss the Drainage design of high rise buildings. [8]



Total No. of Questions : 12]

SEAT No. :

P3033

[Total No. of Pages : 2

[5872]-115

M.E. (Civil - Environmental Engineering)
INDUSTRIAL WASTE WATER TREATMENT AND MANAGEMENT
(2017 Pattern) (Semester - II) (501067)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.
- 2) Figures to the right indicate full marks.
- 3) Your answers will be valued as a whole.
- 4) Assume suitable data, if necessary.
- 5) Use of electronic pocket calculator and steam table is allowed.

Q1) Discuss the Toxicity and Bioassay test. **[5]**

OR

Q2) Give short note on the followings. **[5]**

- a) Equalization and Neutralization.
- b) Flotation and Precipitation.

Q3) Discuss in detail Nitrification and De-nitrification process. **[5]**

OR

Q4) Give detail note on Wet Air Oxidation. **[5]**

Q5) Suggest and comment on the treatment & disposal method of effluent from dairy industry in detail. **[5]**

OR

Q6) Suggest and comment on the treatment & disposal method of effluent from food processing industry. **[5]**

Q7) Enlist the Characteristics and composition of effluent from Mineral Processing Industry. **[5]**

OR

Q8) Discuss how the manufacturing process of Pharma industry is differ than the petroleum industry. **[5]**

P.T.O.

- Q9) a)** What are the common problems encountered in management of CETP. Discuss in detail. [7]
b) Explain with suitable example about the importance of Zero effluent discharge system. [7]

OR

- Q10)a)** What are the general design considerations and principles involve in of CETP? [7]
b) Why it is important to reuse the waste water? How industries play their role in reuse? [7]

- Q11)a)** What is treatment with polymer coagulation? How it is differ from other methods? [8]
b) Enumerate the major design components required for waste water treatment plant for Petroleum industry. Comment on its suitability. [8]

OR

- Q12)a)** Discuss the effluent standards for Dye stuff and dye manufacturing industries. [8]
b) What is an adsorption isotherm? Describe Freundlich adsorption isotherm and Langmuir adsorption isotherm. [8]



Total No. of Questions : 8]

SEAT No. :

P3034

[Total No. of Pages : 2

[5872]-116

M.E.(Civil - Environmental Engineering)

AIR POLLUTION AND CONTROL

(2017 Pattern) (Semester - II) (501068)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Solve Q1 or Q2, Q3 or Q4, Q5 or Q6 and Q7 or Q8.
- 2) Figures to the right indicate full marks.
- 3) Draw neat figures wherever necessary.

Q1) a) What do you understand by stability conditions? Explain its significance in air quality study. **[4]**

b) What is the significance of wind rose diagram in air pollution studies?**[5]**

OR

Q2) a) Draw a figure showing four major layers of the atmosphere along with the temperature profiles. **[4]**

b) A power plant is emitting SO₂ with exhaust rate of 127 g/s through a stack that has effective height of 75m. The atmospheric conditions are moderately to slightly stable. The wind velocity at the top of the stack is 6 m/s. Determine the ground level concentration at a distance of 3 km downwind at the centreline of the plume. $\sigma_y = 280\text{m}$ and $\sigma_z = 170\text{m}$. **[5]**

Q3) a) How does sulphur dioxide is reduced from flue gas? **[4]**

b) Explain in brief about the control of pollution by process modification and change of raw materials. **[5]**

OR

Q4) a) Differentiate between bag house filter and ESP. **[4]**

b) What are the various NO_x control technologies? **[5]**

P.T.O.

- Q5) a)** Write in a tabular form annual and 24 hrs National Ambient Air Quality Standards (NAAQS) for SO_2 , NO_2 , PM_{10} , $\text{PM}_{2.5}$ specified by Central Pollution Control Board. **[8]**
- b)** Explain in brief National Air Quality Monitoring Programme. **[8]**

OR

- Q6) a)** What are modifications in revised Environmental guidelines for siting of Industries? Explain the purpose of providing revised guidelines. **[8]**
- b)** Explain with neat sketch the determination of ambient PM_{10} and $\text{PM}_{2.5}$ using particulate dust sampler. **[8]**
- Q7) a)** What are the causes of indoor air pollution? Explain the measures to be taken to control it. **[8]**
- b)** What are the major air pollutants emitted from vehicular sources? What are the measures taken to control it in Delhi? **[8]**

OR

- Q8) a)** What are the sources of the odours produced in chemical, tanneries, fertilizer and food industries? **[8]**
- b)** What is air cleaning system for control of indoor air pollution? Explain the method of mechanical ventilation. **[8]**



Total No. of Questions : 10]

SEAT No. :

P3035

[Total No. of Pages : 1

[5872]-117

M.E. (Civil - Environmental Engineering)
SOLID WASTE AND HAZARDOUS WASTE MANAGEMENT
(2017 Pattern) (Semester - II) (501069)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Solve question 1 or 2, 3 or 4, 5 or 6, 7 or 8, 9 or 10.
- 2) Each question carries equal marks.
- 3) Figures to the right side indicate full marks.
- 4) Use of calculator is allowed.
- 5) Assume suitable data if necessary.

- Q1)** a) Write a detailed note on physical and chemical properties of MSW. [5]
b) Discuss in detail, Stationary container collection system. [5]

OR

- Q2)** Explain in detail, the functional elements of Solid Waste Management. [10]

- Q3)** Describe in detail, the Thermal technologies and air emissions and its control in detail. Explain the following: (i) Pyrolysis (ii) Refuse derived fuel (iii) Incinerator. [10]

OR

- Q4)** Explain in detail, the different methods of landfilling. [10]

- Q5)** Discuss in detail, magnetic separation and density separation in details. [10]

OR

- Q6)** Explain in detail, the various methods for disposal of Nuclear and Radioactive waste. [10]

- Q7)** Explain in detail about drop off center and buy back center. [10]

OR

- Q8)** What are the different techniques used for handling of Hazardous waste management? Explain in detail. [10]

- Q9)** Explain in detail, the significance of Hazardous waste management. [10]

OR

- Q10)** Write a detailed note on TCLP test. [10]



Total No. of Questions : 8]

SEAT No. :

P3036

[Total No. of Pages : 2

[5872]-118

M.E. (Civil) (Environmental Engineering)
ENVIRONMENTAL IMPACT ASSESSMENT
(2017 Pattern) (Semester - III) (601073)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.*
- 2) *Figures to the right indicate full marks.*
- 3) *Draw neat figures wherever necessary.*
- 4) *Assume necessary data.*
- 5) *Use of scientific calculators is allowed.*

- Q1) a)** What kind of documentation and selection process required for EIA?[4]
- b) What are the different methods used in life cycle assessment. Explain any one method. [5]

OR

- Q2) a)** Explain sensitive area criteria matrices with respect to any construction project. [4]
- b) Explain interaction matrix method. [5]
- Q3) a)** How to identify the type and quantity of air pollutant produced in cement industry. [4]
- b) Explain mitigations for soil and ground water contamination for the landfill site. [5]

OR

- Q4) a)** How to predict of noise levels for any processing industrial areas. [4]
- b) Write different water quality standards and their permissible limits. [5]
- Q5) a)** Explain the role of socioeconomic factors in prediction and assessment of environmental impact? [8]
- b) Write and explain techniques for conflict management and dispute resolution. [8]

OR

P.T.O.

- Q6)** a) Write regulatory requirements for EIA and their importance. [8]
b) Explain resettlement and rehabilitation for any dam project. [8]

- Q7)** a) Write public hearing to be carried out for any express highway. [8]
b) Write and explain general structure of EIA document. [8]

OR

- Q8)** a) Write procedure for environmental Clearance for any industry. [8]
b) Explain post environmental monitoring for any industry. [8]



Total No. of Questions : 8]

SEAT No. :

P3037

[Total No. of Pages : 2

[5872]-119

M.E. (Civil Engg.) (Environmental Engineering)

RESEARCH METHODOLOGY

(2017 Pattern) (Semester-III) (601074)

Time : 3 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2; Q.3 or Q.4; Q.5 or Q.6: and Q.7 or Q.8.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary and clearly state.*

- Q1)** a) Enlist steps involved in a Research Process. [5]
b) Write short note on different sources of literature. [4]

OR

- Q2)** a) Define Research and describe any two types of research. [5]
b) Explain the importance of review of literature in Research. [4]

- Q3)** a) Explain the meaning of analysis of variance. Describe briefly the technique of analysis of variance. [5]
b) Explain the criteria of goodness of a measurement scale. [4]

OR

- Q4)** a) Explain the different types of report. [5]
b) Distinguish factor analysis and cluster analysis [4]

- Q5)** a) What are the primary sources of data collection and explain any two?[8]
b) Distinguish between Research Methods and Research Methodology.[8]

OR

P.T.O.

- Q6)** a) What are the merits and demerits of collection of data through questionnaires? [8]
b) Define Multi-dimensional Scaling (MDS). Differentiate metric & non-metric approach of MDS with an example. [8]
- Q7)** a) Explain various steps involved in writing a research paper also explain how to search journal for paper publication. [7]
b) State how to check plagiarism? Justify with an example of any one tool for plagiarism checking. [5]
c) Explain procedure of Obtaining patents. [4]

OR

- Q8)** a) State the need of research report writing and state various steps involved in writing a report. [7]
b) Enlist different factors to be considered for effective presentation with an example. [5]
c) List out the steps involved in publishing a research article in journal. [4]

